

Solution:

$$\mathbf{E}(t) = \operatorname{Re}\{\hat{\mathbf{x}} E_0 e^{-jkz} e^{j\omega t}\} = \hat{\mathbf{x}} E_0 \cos(\omega t - kz)$$

$$\mathbf{H}(t) = \operatorname{Re}\{\hat{\mathbf{y}} \frac{k}{\omega\mu} E_0 e^{-jkz} e^{j\omega t}\} = \hat{\mathbf{y}} \frac{k}{\omega\mu} E_0 \cos(\omega t - kz)$$

$$? \quad \mathbf{S}(t) = \mathbf{E} \times \mathbf{H} = \hat{\mathbf{z}} \frac{k}{\omega\mu} E_0^2 \cos^2(\omega t - kz)$$

$$\langle \mathbf{S} \rangle = \frac{1}{2} \operatorname{Re}\{\mathbf{E} \times \mathbf{H}^*\} = \hat{\mathbf{z}} \frac{k}{2\omega\mu} E_0^2$$

$$U_E = \frac{\epsilon E_0^2}{2} \cos^2(\omega t - kz)$$

$$U_H = \frac{k^2 E_0^2}{2\omega^2 \mu} \cos^2(\omega t - kz)$$

$$\langle U_E \rangle = \frac{\epsilon}{4} E_0^2$$

$$\langle U_H \rangle = \frac{k^2}{4\omega^2 \mu} E_0^2$$

Problems

2.1 Let $\mathbf{A} = 5\hat{\mathbf{x}} + 6yz\hat{\mathbf{y}} + x^3\hat{\mathbf{z}}$; find $\nabla \times \mathbf{A}$ and $\nabla \cdot \mathbf{A}$.

2.2 Let $\phi = xyz$; find $\nabla\phi$ and $\nabla \cdot \nabla\phi$.

2.3 Let $\mathbf{a} = a_1\hat{\mathbf{x}} + a_2\hat{\mathbf{y}} + a_3\hat{\mathbf{z}}$ and $\mathbf{b} = b_1\hat{\mathbf{x}} + b_2\hat{\mathbf{y}} + b_3\hat{\mathbf{z}}$. Show that equations (2.9), (2.11a), and (2.12) are true.

2.4 Show that $\nabla \times (\mathbf{a} + \mathbf{b}) = \nabla \times \mathbf{a} + \nabla \times \mathbf{b}$ and $\nabla \cdot (\mathbf{a} + \mathbf{b}) = \nabla \cdot \mathbf{a} + \nabla \cdot \mathbf{b}$.

2.5 Show that $\nabla(\phi_1 \phi_2) = \phi_1 \nabla \phi_2 + \phi_2 \nabla \phi_1$ and that $\nabla \cdot (\phi \mathbf{A}) = \mathbf{A} \cdot \nabla \phi + \phi \nabla \cdot \mathbf{A}$.

2.6 Show that $\nabla \times (\phi \mathbf{A}) = \nabla \phi \times \mathbf{A} + \phi \nabla \times \mathbf{A}$.

2.7 In a source-free region, $\mathbf{H} = z\hat{\mathbf{y}} + y\hat{\mathbf{z}}$. Does $\mathbf{D}$ vary with time?

2.8 What is the charge density in a region where $\mathbf{D} = 2x\hat{\mathbf{x}}$?

2.9 Find the magnetic field $\mathbf{B}(y,t)$ associated with the electric field $\mathbf{E}(y,t)$ given as follows:

$$\mathbf{E}(y,t) = \hat{\mathbf{x}} 0.3 \cos(\omega t + ky)$$

where ω and k are constants.

2.10 Express k in terms of the magnetic permeability and dielectric permittivity of the medium when the electromagnetic fields are given in Problem 2.9 in a source-free region.

2.11 Let $\mathbf{E}_1$, $\mathbf{B}_1$, $\mathbf{H}_1$, and $\mathbf{D}_1$ satisfy equations (2.1)–(2.4) with given $\mathbf{J}_1$ and ρ_{v1} . Let also $\mathbf{E}_2$, $\mathbf{B}_2$, $\mathbf{H}_2$, and $\mathbf{D}_2$ satisfy equations (2.1)–(2.4) with given $\mathbf{J}_2$ and ρ_{v2} . What are the electromagnetic fields due to a current $\mathbf{J}_t$ and charge ρ_{vt} where $\mathbf{J}_t = \mathbf{J}_1 + \mathbf{J}_2$ and $\rho_{vt} = \rho_{v1} + \rho_{v2}$? You must show that your proposed solution satisfies Maxwell's equations. What is the appropriate name for the theorem you have just proved?

- 2.12 (a) It is known that the vector $\mathbf{a}$ is equal to zero at one point. Does that imply that $\nabla \times \mathbf{a} = 0$ at that point? Give a counter-example if your answer is no.
 (b) Does $\mathbf{E} = 0$ on a line always imply $\nabla \times \mathbf{E} = 0$ on that line? Give a counter-example if the answer is no.
 (c) It is found that the $\mathbf{E}$ field is zero on a surface. Does it follow that $\partial \mathbf{B} / \partial t = 0$ on that surface?
- 2.13 Show that equations (2.22c) and (2.22d) can be derived from equations (2.22a), (2.22b), and the conservation equation (2.23).
- 2.14 To represent time-harmonic fields, most physics books use the factor $e^{-i\omega t}$ instead of $e^{j\omega t}$, which most electrical engineering books use. For a time-harmonic real function $A(x, y, z, t) = a(x, y, z) \cos(\omega t + \phi)$, find the phasor notation that corresponds to the physicists' convention. What is the corresponding conversion rule by which phasors can be transformed back to the real-time expression?
- 2.15 Refer to Problem 2.14 about the notation $e^{-i\omega t}$ adopted in most physics books. Write the time-harmonic Maxwell's equations using that notation.
- 2.16 What is the range of effective permittivity of the ionosphere at AM broadcasting frequencies? Use the following data: $N = 10^{12} \text{ m}^{-3}$ and $f = 500 \text{ kHz}$ to 1 MHz .
- 2.17 Show that the dimension of each term of equation (2.36) is watts per cubic meter.
- 2.18 Indicate in watts, meters, and joules the dimensions of the following quantities: (a) $\mathbf{E} \cdot \mathbf{D}$, (b) $\mathbf{H} \cdot \mathbf{B}$, and (c) $\mathbf{S}$.
- 2.19 Let $\mathbf{E} = (\hat{x} + j\hat{y})e^{jz}$ and $\mathbf{H} = (j\hat{y} - \hat{x})e^{-jz}$. Find $\mathbf{S}$ in terms of z and ωt and find $\langle \mathbf{S} \rangle$.
- 2.20 Show that $\mathbf{S} \neq \text{Re} \{ \mathbf{E} \times \mathbf{H} e^{j\omega t} \}$.
- 2.21 Show that $\mathbf{S} \neq \text{Re} \{ \mathbf{E} e^{j\omega t} \times \mathbf{H} e^{j\omega t} \}$.
- 2.22 Compare the energy stored in a cubic region one meter on a side which has a uniform $\mathbf{E}$ field of 10^4 V/m to the energy stored in a similar region with a uniform $\mathbf{B}$ field of 10^4 G . (One $\text{G} = 10^{-4} \text{ Wb/m}^2$). The medium is air.
- 2.23 Repeat Problem 2.22 for the case where the medium is water instead of air. Use $\epsilon = 80 \epsilon_0$ and $\mu = \mu_0$ for water.

CHAPTER 2

2.1 $\nabla \times \vec{A} = -6y \hat{x} - 3x^2 \hat{y}, \quad \nabla \cdot \vec{A} = 6z$

2.2 $\nabla \phi = yz \hat{x} + xz \hat{y} + xy \hat{z}, \quad \nabla \cdot \nabla \phi = 0$

2.3 $\vec{a} = a_1 \hat{x} + a_2 \hat{y} + a_3 \hat{z}, \quad \vec{b} = b_1 \hat{x} + b_2 \hat{y} + b_3 \hat{z}$

$$\nabla \times \vec{a} = \hat{x} \left(\frac{\partial a_3}{\partial y} - \frac{\partial a_2}{\partial z} \right) + \hat{y} \left(\frac{\partial a_1}{\partial z} - \frac{\partial a_3}{\partial x} \right) + \hat{z} \left(\frac{\partial a_2}{\partial x} - \frac{\partial a_1}{\partial y} \right)$$

$$\nabla \times (\nabla \times \vec{a}) = \hat{x} \left(-\frac{\partial^2 a_1}{\partial y^2} + \frac{\partial^2 a_2}{\partial y \partial z} - \frac{\partial^2 a_3}{\partial z^2} + \frac{\partial^2 a_3}{\partial z \partial x} \right) + \hat{y} \left(\frac{\partial^2 a_3}{\partial y \partial z} - \frac{\partial^2 a_2}{\partial z^2} + \frac{\partial^2 a_1}{\partial z \partial y} - \frac{\partial^2 a_1}{\partial x^2} \right) + \hat{z} \left(-\frac{\partial^2 a_2}{\partial y^2} + \frac{\partial^2 a_1}{\partial y \partial z} + \frac{\partial^2 a_1}{\partial x \partial y} - \frac{\partial^2 a_3}{\partial x^2} \right)$$

$$\nabla \cdot \vec{a} = \frac{\partial a_1}{\partial x} + \frac{\partial a_2}{\partial y} + \frac{\partial a_3}{\partial z}, \quad \nabla \cdot (\nabla \times \vec{a}) = \hat{x} \left(\frac{\partial^2 a_1}{\partial x^2} + \frac{\partial^2 a_2}{\partial x \partial y} + \frac{\partial^2 a_3}{\partial x \partial z} \right) + \hat{y} \left(\frac{\partial^2 a_1}{\partial x \partial y} + \frac{\partial^2 a_2}{\partial y^2} + \frac{\partial^2 a_3}{\partial y \partial z} \right) + \hat{z} \left(\frac{\partial^2 a_1}{\partial x \partial z} + \frac{\partial^2 a_2}{\partial y \partial z} + \frac{\partial^2 a_3}{\partial z^2} \right)$$

$$\nabla^2 \vec{a} = \hat{x} \left(\frac{\partial^2 a_1}{\partial x^2} + \frac{\partial^2 a_1}{\partial y^2} + \frac{\partial^2 a_1}{\partial z^2} \right) + \hat{y} \left(\frac{\partial^2 a_2}{\partial x^2} + \frac{\partial^2 a_2}{\partial y^2} + \frac{\partial^2 a_2}{\partial z^2} \right) + \hat{z} \left(\frac{\partial^2 a_3}{\partial x^2} + \frac{\partial^2 a_3}{\partial y^2} + \frac{\partial^2 a_3}{\partial z^2} \right)$$

$$\therefore \nabla \times (\nabla \times \vec{a}) = \nabla (\nabla \cdot \vec{a}) - \nabla^2 \vec{a}$$

$$\nabla \cdot (\nabla \times \vec{a}) = \frac{\partial^2 a_3}{\partial x \partial y} - \frac{\partial^2 a_1}{\partial x \partial z} + \frac{\partial^2 a_1}{\partial y \partial z} - \frac{\partial^2 a_3}{\partial x \partial y} + \frac{\partial^2 a_2}{\partial x \partial z} - \frac{\partial^2 a_1}{\partial y \partial z} = 0$$

$$\vec{a} \times \vec{b} = \hat{x} (a_2 b_3 - a_3 b_2) + \hat{y} (a_3 b_1 - a_1 b_3) + \hat{z} (a_1 b_2 - a_2 b_1)$$

$$\nabla \cdot (\vec{a} \times \vec{b}) = \frac{\partial}{\partial x} (a_2 b_3 - a_3 b_2) + \frac{\partial}{\partial y} (a_3 b_1 - a_1 b_3) + \frac{\partial}{\partial z} (a_1 b_2 - a_2 b_1) = a_2 \left(\frac{\partial b_3}{\partial x} - \frac{\partial b_1}{\partial z} \right) + b_3 \left(\frac{\partial a_2}{\partial x} - \frac{\partial a_1}{\partial z} \right) - a_3 \left(\frac{\partial b_1}{\partial y} - \frac{\partial b_2}{\partial z} \right) + b_2 \left(\frac{\partial a_1}{\partial y} - \frac{\partial a_3}{\partial z} \right) + b_1 \left(\frac{\partial a_3}{\partial y} - \frac{\partial a_2}{\partial z} \right) + a_1 \left(\frac{\partial b_2}{\partial z} - \frac{\partial b_3}{\partial y} \right)$$

$$\vec{b} \cdot (\nabla \times \vec{a}) = b_1 \left(\frac{\partial a_3}{\partial y} - \frac{\partial a_2}{\partial z} \right) + b_2 \left(\frac{\partial a_1}{\partial z} - \frac{\partial a_3}{\partial x} \right) + b_3 \left(\frac{\partial a_2}{\partial x} - \frac{\partial a_1}{\partial y} \right)$$

$$\vec{a} \cdot (\nabla \times \vec{b}) = a_1 \left(\frac{\partial b_2}{\partial z} - \frac{\partial b_3}{\partial y} \right) + a_2 \left(\frac{\partial b_1}{\partial z} - \frac{\partial b_3}{\partial x} \right) + a_3 \left(\frac{\partial b_1}{\partial x} - \frac{\partial b_2}{\partial y} \right)$$

$$\therefore \nabla \cdot (\vec{a} \times \vec{b}) = \vec{b} \cdot \nabla \times \vec{a} - \vec{a} \cdot \nabla \times \vec{b}$$

2.4 Let $\vec{a} = a_1 \hat{x} + a_2 \hat{y} + a_3 \hat{z}, \quad \vec{b} = b_1 \hat{x} + b_2 \hat{y} + b_3 \hat{z}$

$$\nabla \times (\vec{a} + \vec{b}) = \nabla \times [(a_1 + b_1) \hat{x} + (a_2 + b_2) \hat{y} + (a_3 + b_3) \hat{z}] = \hat{x} \left[\frac{\partial}{\partial y} (a_3 + b_3) - \frac{\partial}{\partial z} (a_2 + b_2) \right] + \hat{y} \left[\frac{\partial}{\partial z} (a_1 + b_1) - \frac{\partial}{\partial x} (a_3 + b_3) \right] + \hat{z} \left[\frac{\partial}{\partial x} (a_2 + b_2) - \frac{\partial}{\partial y} (a_1 + b_1) \right]$$

$$\nabla \times \vec{a} = \hat{x} \left(\frac{\partial a_3}{\partial y} - \frac{\partial a_2}{\partial z} \right) + \hat{y} \left(\frac{\partial a_1}{\partial z} - \frac{\partial a_3}{\partial x} \right) + \hat{z} \left(\frac{\partial a_2}{\partial x} - \frac{\partial a_1}{\partial y} \right)$$

$$\nabla \times \vec{b} = \hat{x} \left(\frac{\partial b_3}{\partial y} - \frac{\partial b_2}{\partial z} \right) + \hat{y} \left(\frac{\partial b_1}{\partial z} - \frac{\partial b_3}{\partial x} \right) + \hat{z} \left(\frac{\partial b_2}{\partial x} - \frac{\partial b_1}{\partial y} \right)$$

$$\therefore \nabla \times (\vec{a} + \vec{b}) = \nabla \times \vec{a} + \nabla \times \vec{b}$$

$$\nabla \cdot (\vec{a} + \vec{b}) = \frac{\partial}{\partial x} (a_1 + b_1) + \frac{\partial}{\partial y} (a_2 + b_2) + \frac{\partial}{\partial z} (a_3 + b_3) = \left(\frac{\partial a_1}{\partial x} + \frac{\partial a_2}{\partial y} + \frac{\partial a_3}{\partial z} \right) + \left(\frac{\partial b_1}{\partial x} + \frac{\partial b_2}{\partial y} + \frac{\partial b_3}{\partial z} \right) = \nabla \cdot \vec{a} + \nabla \cdot \vec{b}$$

2.5 $\nabla (\phi_1 \phi_2) = \hat{x} \frac{\partial}{\partial x} (\phi_1 \phi_2) + \hat{y} \frac{\partial}{\partial y} (\phi_1 \phi_2) + \hat{z} \frac{\partial}{\partial z} (\phi_1 \phi_2) = \hat{x} \left[\phi_1 \frac{\partial \phi_2}{\partial x} + \phi_2 \frac{\partial \phi_1}{\partial x} \right] + \hat{y} \left[\phi_1 \frac{\partial \phi_2}{\partial y} + \phi_2 \frac{\partial \phi_1}{\partial y} \right] + \hat{z} \left[\phi_1 \frac{\partial \phi_2}{\partial z} + \phi_2 \frac{\partial \phi_1}{\partial z} \right] = \phi_1 \left[\hat{x} \frac{\partial \phi_2}{\partial x} + \hat{y} \frac{\partial \phi_2}{\partial y} + \hat{z} \frac{\partial \phi_2}{\partial z} \right] + \phi_2 \left[\hat{x} \frac{\partial \phi_1}{\partial x} + \hat{y} \frac{\partial \phi_1}{\partial y} + \hat{z} \frac{\partial \phi_1}{\partial z} \right] = \phi_1 \nabla \phi_2 + \phi_2 \nabla \phi_1$

Let $\vec{A} = \hat{x} A_x + \hat{y} A_y + \hat{z} A_z$

$$\nabla \cdot (\phi \vec{A}) = \frac{\partial}{\partial x} (\phi A_x) + \frac{\partial}{\partial y} (\phi A_y) + \frac{\partial}{\partial z} (\phi A_z) = \phi \left[\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \right] + \left[\frac{\partial \phi}{\partial x} A_x + \frac{\partial \phi}{\partial y} A_y + \frac{\partial \phi}{\partial z} A_z \right] = \phi \nabla \cdot \vec{A} + \nabla \phi \cdot \vec{A}$$

2.6 $\nabla \times (\phi \vec{A}) = \hat{x} \left[\frac{\partial}{\partial y} (\phi A_z) - \frac{\partial}{\partial z} (\phi A_y) \right] + \hat{y} \left[\frac{\partial}{\partial z} (\phi A_x) - \frac{\partial}{\partial x} (\phi A_z) \right] + \hat{z} \left[\frac{\partial}{\partial x} (\phi A_y) - \frac{\partial}{\partial y} (\phi A_x) \right]$
 $= \phi \left[\hat{x} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \hat{y} \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \hat{z} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \right] + \hat{x} \left(A_z \frac{\partial \phi}{\partial y} - A_y \frac{\partial \phi}{\partial z} \right) + \hat{y} \left(A_x \frac{\partial \phi}{\partial z} - A_z \frac{\partial \phi}{\partial x} \right) + \hat{z} \left(A_y \frac{\partial \phi}{\partial x} - A_x \frac{\partial \phi}{\partial y} \right)$
 $= \phi \nabla \times \vec{A} + \nabla \phi \times \vec{A}$

2.7 $\nabla \times \vec{H} = \nabla \times (\partial \hat{y} + y \hat{z}) = \hat{x} (1 - 1) = 0 \Rightarrow \frac{\partial \vec{D}}{\partial t} = 0$

$\therefore \vec{D}$ is not varying with time.

2.8 $\nabla \cdot \bar{D} = \nabla \cdot (2x\hat{x}) = 2 \Rightarrow \rho_v = 2 \text{ C/m}^3$

2.9 $\bar{E} = 0.3 e^{jk_y y} \hat{x}$

$$\nabla \times \bar{E} = -\hat{z} (0.3 j k e^{jk_y y}) = -j\omega \bar{B}$$

$$\bar{B} = 0.3 \frac{k}{\omega} e^{jk_y y} \hat{z}$$

$$\bar{B}(y,t) = \frac{0.3k}{\omega} \cos(\omega t + k_y y) \hat{z}$$

2.10 $\nabla \times \bar{H} = \bar{J} + \frac{\partial \bar{D}}{\partial t}$, substituting $\bar{H} = \frac{1}{\mu} \bar{B} = \frac{0.3k}{\omega\mu} \cos(\omega t + k_y y) \hat{z}$,

$\bar{J} = 0$, $\bar{D} = \epsilon \bar{E} = 0.3\epsilon \cos(\omega t + k_y y) \hat{x}$ into this equation yields:

$$k^2 = \omega^2 \mu \epsilon \text{ or } k = \omega \sqrt{\mu \epsilon}.$$

2.11 $\nabla \times \bar{E}_1 = -\frac{\partial \bar{B}_1}{\partial t}$, $\nabla \times \bar{H}_1 = \bar{J}_1 + \frac{\partial \bar{D}_1}{\partial t}$, $\nabla \cdot \bar{B}_1 = 0$, $\nabla \cdot \bar{D}_1 = \rho_{v1}$

$\nabla \times \bar{E}_2 = -\frac{\partial \bar{B}_2}{\partial t}$, $\nabla \times \bar{H}_2 = \bar{J}_2 + \frac{\partial \bar{D}_2}{\partial t}$, $\nabla \cdot \bar{B}_2 = 0$, $\nabla \cdot \bar{D}_2 = \rho_{v2}$

$$\nabla \times \bar{E}_t = \nabla \times (\bar{E}_1 + \bar{E}_2) = \nabla \times \bar{E}_1 + \nabla \times \bar{E}_2 = -\frac{\partial \bar{B}_1}{\partial t} - \frac{\partial \bar{B}_2}{\partial t} = -\frac{\partial (\bar{B}_1 + \bar{B}_2)}{\partial t} = -\frac{\partial \bar{B}_t}{\partial t}$$

$$\nabla \times \bar{H}_t = \nabla \times (\bar{H}_1 + \bar{H}_2) = \nabla \times \bar{H}_1 + \nabla \times \bar{H}_2 = \bar{J}_1 + \frac{\partial \bar{D}_1}{\partial t} + \bar{J}_2 + \frac{\partial \bar{D}_2}{\partial t} = (\bar{J}_1 + \bar{J}_2) + \frac{\partial (\bar{D}_1 + \bar{D}_2)}{\partial t} = \bar{J}_t + \frac{\partial \bar{D}_t}{\partial t}$$

$$\nabla \cdot \bar{B}_t = \nabla \cdot (\bar{B}_1 + \bar{B}_2) = \nabla \cdot \bar{B}_1 + \nabla \cdot \bar{B}_2 = 0 + 0 = 0$$

$$\nabla \cdot \bar{D}_t = \nabla \cdot (\bar{D}_1 + \bar{D}_2) = \nabla \cdot \bar{D}_1 + \nabla \cdot \bar{D}_2 = \rho_{v1} + \rho_{v2} = \rho_{vt}$$

$\therefore \bar{E}_t = \bar{E}_1 + \bar{E}_2$, $\bar{H}_t = \bar{H}_1 + \bar{H}_2$, $\bar{B}_t = \bar{B}_1 + \bar{B}_2$ and $\bar{D}_t = \bar{D}_1 + \bar{D}_2$ satisfy Maxwell's equations with sources $\bar{J}_t = \bar{J}_1 + \bar{J}_2$ and $\rho_{vt} = \rho_{v1} + \rho_{v2}$.

This is the superposition theorem.

2.12 (a) No. $\bar{a} = \hat{x} \sin z + \hat{y} \sin x + \hat{z} \sin y = 0$ at $(0,0,0)$, but $\nabla \times \bar{a} = \hat{x} \cos y + \hat{y} \cos z + \hat{z} \cos x$ at $(0,0,0)$.

(b) No. $\bar{E} = \hat{x} \sin z + \hat{y} \sin x = 0$ on y -axis, but $\nabla \times \bar{E} = \hat{y} \cos z + \hat{z} \cos x \neq 0$ on y -axis.

(c) NO. $\bar{E} = \hat{x} \sin z = 0$ on $z=0$ plane, but $\nabla \times \bar{E} = \hat{y} \cos z \neq 0$ on $z=0$ plane $\Rightarrow \frac{\partial \bar{B}}{\partial t} \neq 0$ on that plane.

2.13 $\nabla \cdot (\nabla \times \bar{E}) \equiv 0 \Rightarrow j\omega \nabla \cdot \bar{B} = 0 \Rightarrow \nabla \cdot \bar{B} = 0$

$$\nabla \cdot (\nabla \times \bar{H}) \equiv 0 \Rightarrow \nabla \cdot \bar{J} + j\omega \nabla \cdot \bar{D} = 0 \Rightarrow j\omega (-\rho_v + \nabla \cdot \bar{D}) = 0 \Rightarrow \nabla \cdot \bar{D} = \rho_v$$

2.14 $A(x,y,z,t) = a(x,y,z) \cos(\omega t + \phi) = a(x,y,z) \cos(-\omega t - \phi)$

$$e^{j\omega t}: A(x,y,z,t) = \text{Re}\{a(x,y,z) e^{j\phi} e^{j\omega t}\} \longleftrightarrow a(x,y,z) e^{j\phi}$$

$$e^{-j\omega t}: A(x,y,z,t) = \text{Re}\{a(x,y,z) e^{-j\phi} e^{-j\omega t}\} \longleftrightarrow a(x,y,z) e^{-j\phi}$$

2.15 $\nabla \times \bar{E} = j\omega \bar{B}$, $\nabla \times \bar{H} = \bar{J} - j\omega \bar{D}$, $\nabla \cdot \bar{B} = 0$, $\nabla \cdot \bar{D} = \rho_v$

2.16 $\epsilon = \epsilon_0 (1 - \frac{\omega_p^2}{\omega^2})$ $\omega_p = \sqrt{\frac{N e^2}{m \epsilon_0}} = \left[\frac{10^{12} \times (1.6 \times 10^{-19})^2}{9.11 \times 10^{-31}} \times 36 \pi \times 10^9 \right]^{1/2} = 5.64 \times 10^7 \text{ Hz}$

at 500 kHz : $\epsilon = \epsilon_0 \left[1 - \left(\frac{5.64 \times 10^7}{2\pi \times 5 \times 10^5} \right)^2 \right] = -323 \epsilon_0$

at 1 MHz : $\epsilon = \epsilon_0 \left[1 - \left(\frac{5.64 \times 10^7}{2\pi \times 10^6} \right)^2 \right] = -80.43 \epsilon_0$

$$\begin{aligned}
 2.17 \quad [\vec{E}] &= \frac{V}{m}, [\vec{H}] = \frac{A}{m} \therefore [\nabla \cdot (\vec{E} \times \vec{H})] = \frac{V \cdot A}{m^3} = \frac{\text{Watt}}{m^3} \\
 [\mu \vec{H} \cdot \vec{H}] &= \frac{V \cdot \text{sec}}{m^2} \cdot \frac{A}{m} = \frac{\text{Watt} \cdot \text{sec}}{m^3}, \therefore \left[\frac{\partial}{\partial t} \left(\frac{1}{2} \mu \vec{H} \cdot \vec{H} \right) \right] = \frac{\text{Watt}}{m^3} \\
 [\epsilon \vec{E} \cdot \vec{E}] &= \frac{C}{m^2} \cdot \frac{V}{m} = \frac{\text{Joule}}{m^3}, \therefore \left[\frac{\partial}{\partial t} \left(\frac{1}{2} \epsilon \vec{E} \cdot \vec{E} \right) \right] = \frac{\text{Watt}}{m^3} \\
 [\vec{J} \cdot \vec{E}] &= \frac{A}{m^2} \cdot \frac{V}{m} = \frac{\text{Watt}}{m^3}
 \end{aligned}$$

$$\begin{aligned}
 2.18 \quad (a) \quad [\vec{E} \cdot \vec{D}] &= \frac{V}{m} \cdot \frac{C}{m^2} = \frac{\text{Joule}}{m^3} \\
 (b) \quad [\vec{H} \cdot \vec{B}] &= \frac{A}{m} \cdot \frac{V \cdot \text{sec}}{m^2} = \frac{\text{Watt} \cdot \text{sec}}{m^3} = \frac{\text{Joule}}{m^3} \\
 (c) \quad [\vec{S}] = [\vec{E} \times \vec{H}] &= \frac{V}{m} \cdot \frac{A}{m} = \frac{\text{Watt}}{m^2}
 \end{aligned}$$

$$\begin{aligned}
 2.19 \quad \vec{E} &= (\hat{x} + j\hat{y})e^{jz} \longleftrightarrow \vec{E}(t) = \hat{x} \cos(\omega t - z) - \hat{y} \sin(\omega t - z) \\
 \vec{H} &= (\hat{y} - j\hat{x})e^{jz} \longleftrightarrow \vec{H}(t) = \hat{y} \cos(\omega t - z) + \hat{x} \sin(\omega t - z) \\
 \vec{S} &= \vec{E} \times \vec{H} = \hat{z} [\cos^2(\omega t - z) + \sin^2(\omega t - z)] = \hat{z} \\
 \vec{S} &= \vec{E} \times \vec{H}^* = (\hat{x} + j\hat{y}) \times (\hat{y} + j\hat{x}) = \hat{z} 2 \Rightarrow \langle \vec{S} \rangle = \frac{1}{2} \text{Re} \{ \vec{S} \} = \hat{z}
 \end{aligned}$$

$$\begin{aligned}
 2.20 \quad \text{Let } \vec{E} &= \vec{E}_R + j\vec{E}_I \text{ and } \vec{H} = \vec{H}_R + j\vec{H}_I \\
 \vec{E} &= \text{Re} \{ \vec{E} e^{j\omega t} \} = \vec{E}_R \cos \omega t - \vec{E}_I \sin \omega t, \vec{H} = \text{Re} \{ \vec{H} e^{j\omega t} \} = \vec{H}_R \cos \omega t - \vec{H}_I \sin \omega t \\
 \vec{S} &= \vec{E} \times \vec{H} = (\vec{E}_R \times \vec{H}_R) \cos^2 \omega t + (\vec{E}_I \times \vec{H}_I) \sin^2 \omega t - (\vec{E}_R \times \vec{H}_I + \vec{E}_I \times \vec{H}_R) \sin \omega t \cos \omega t \\
 \text{But } \vec{H} e^{j\omega t} &= (\vec{H}_R \cos \omega t - \vec{H}_I \sin \omega t) + j(\vec{H}_R \sin \omega t + \vec{H}_I \cos \omega t) \text{ and} \\
 \text{Re} \{ \vec{E} \times \vec{H} e^{j\omega t} \} &= \vec{E}_R \times \vec{H}_R \cos \omega t - \vec{E}_R \times \vec{H}_I \sin \omega t - \vec{E}_I \times \vec{H}_R \sin \omega t - \vec{E}_I \times \vec{H}_I \cos \omega t \neq \vec{S}
 \end{aligned}$$

$$\begin{aligned}
 2.21 \quad \text{Let } \vec{E} &= \vec{E}_R + j\vec{E}_I \text{ and } \vec{H} = \vec{H}_R + j\vec{H}_I \\
 \vec{E} e^{j\omega t} &= (\vec{E}_R \cos \omega t - \vec{E}_I \sin \omega t) + j(\vec{E}_R \sin \omega t + \vec{E}_I \cos \omega t) \\
 \vec{H} e^{j\omega t} &= (\vec{H}_R \cos \omega t - \vec{H}_I \sin \omega t) + j(\vec{H}_R \sin \omega t + \vec{H}_I \cos \omega t) \\
 \text{Re} \{ \vec{E} e^{j\omega t} \times \vec{H} e^{j\omega t} \} &= (\vec{E}_R \times \vec{H}_R) \cos^2 \omega t + (\vec{E}_I \times \vec{H}_I) \sin^2 \omega t - (\vec{E}_R \times \vec{H}_I + \vec{E}_I \times \vec{H}_R) \sin \omega t \cos \omega t \\
 &\quad - (\vec{E}_I \times \vec{H}_R) \cos^2 \omega t - (\vec{E}_R \times \vec{H}_R) \sin^2 \omega t - (\vec{E}_R \times \vec{H}_I + \vec{E}_I \times \vec{H}_R) \sin \omega t \cos \omega t \\
 &= (\vec{E}_R \times \vec{H}_R - \vec{E}_I \times \vec{H}_I) (\cos^2 \omega t - \sin^2 \omega t) - 2(\vec{E}_R \times \vec{H}_I + \vec{E}_I \times \vec{H}_R) \sin \omega t \cos \omega t \\
 \therefore \vec{S} &\neq \text{Re} \{ \vec{E} e^{j\omega t} \times \vec{H} e^{j\omega t} \}
 \end{aligned}$$

$$\begin{aligned}
 2.22 \quad U_e &= \frac{1}{2} \epsilon_0 \vec{E} \cdot \vec{E} = \frac{1}{2} \times \frac{1}{36\pi} \times 10^{-9} \times (10^4)^2 = 4.42 \times 10^{-4} \text{ (Joule/m}^3\text{)} \\
 U_h &= \frac{1}{2} \mu_0 \vec{H} \cdot \vec{H} = \frac{1}{2} \times \frac{1}{4\pi} \times 10^{-7} \times (10^4 \times 10^{-4})^2 = 3.98 \times 10^{-5} \text{ (Joule/m}^3\text{)} \\
 \therefore U_h/U_e &= 9 \times 10^8
 \end{aligned}$$

$$\begin{aligned}
 2.23 \quad U_e &= \frac{1}{2} \epsilon \vec{E} \cdot \vec{E} = \frac{1}{2} \times \frac{80}{36\pi} \times 10^{-9} \times (10^4)^2 = 3.54 \times 10^{-2} \text{ (Joule/m}^3\text{)} \\
 U_h &= 3.98 \times 10^{-5} \text{ (Joule/m}^3\text{)} \\
 \therefore U_h/U_e &= 1.13 \times 10^{-7}
 \end{aligned}$$